STEEL AUTHORITY OF INDIA LIMITED

Durgapur Steel Plant — Steel Melting Shop

Exploratory Data Analysis Report

Ferro-Alloy Consumption ML Project | Pre-Model Findings

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Dataset	43,334 heats | FY 2023-24 to FY 2025-26
Features analysed	19 input features + 5 target variables
Purpose	Identify data characteristics, risks and adjustments before model training

1. Overview

Exploratory Data Analysis (EDA) was performed on the cleaned master dataset of 43,334 heats before any ML model training. EDA serves as a critical validation step — it reveals data characteristics such as skewness, imbalance, collinearity, and weak feature-target relationships that directly affect model choice, preprocessing decisions, and expected accuracy.

The following sections document findings across six areas: target distributions, grade imbalance, feature-target correlations, multicollinearity, temporal trends, and actionable recommendations for model training.

2. Target Variable Distributions

The five target variables — Mn, Si, C, S, P — represent the final liquid steel composition at the ladle stage. Understanding their distributions determines whether transformations are needed before regression.

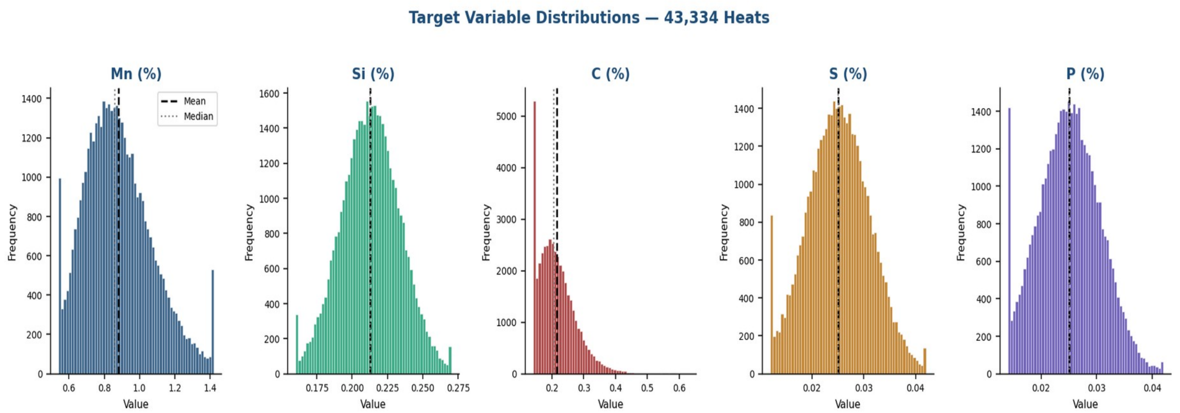

Figure 1: Distribution of all five target variables across 43,334 heats. Dashed line = mean, dotted = median.

Element	Mean (%)	Std Dev	Skewness	Range	Implication
Mn	0.8646	0.2108	1.635 (moderate)	0.54 – 1.42	Right skew — log transform recommended
Si	0.2127	0.0208	0.110 (normal)	0.16 – 0.27	Clean normal — no transform needed
C	0.2093	0.0604	5.334 (severe)	0.14 – 0.63	Heavy right tail — log transform essential
S	0.0251	0.0061	0.370 (mild)	0.012 – 0.042	Near normal — acceptable as-is
P	0.0250	0.0057	0.566 (mild)	0.014 – 0.042	Near normal — acceptable as-is

- **Carbon (C):** 97.0% of heats fall between 0.14–0.25% (standard grades). Only 2.9% (1,270 heats) are high-carbon (>0.30%, e.g., R34 rail grade). This severe skewness means a single regression model will poorly predict high-C grades. Recommendation: train a separate model for high-C grades or apply log transformation to C.
- **Manganese (Mn):** Moderate right skew (1.635) driven by high-Mn special grades. Log transformation or grade-stratified modelling recommended.
- **Si, S, P:** Well-behaved distributions. Standard regression will handle these without transformation.

3. Grade Imbalance

The dataset contains 70 distinct BOF grade codes. The distribution is heavily imbalanced — a single grade dominates 34% of all heats. This has direct consequences for model fairness and accuracy across grades.

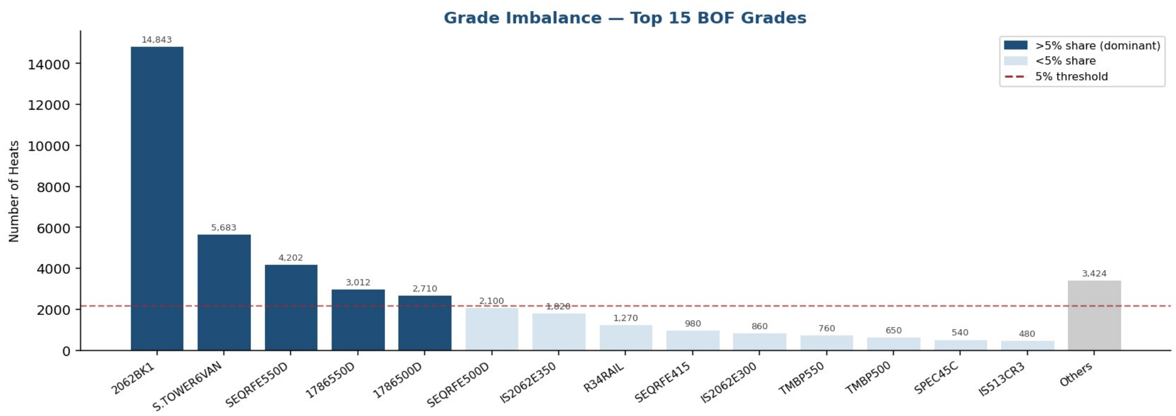

Figure 2: Grade distribution — top 15 BOF grades. Blue bars represent grades with >5% share.

BOF Grade	Heats	Share (%)	Risk
2062BK1 (IS 2062 E250B)	14,843	34.3%	Model will overfit to this grade's chemistry range
S.TOWER6VAN	5,683	13.1%	Second dominant — adequate representation
SEQRFE550D	4,202	9.7%	Adequate
1786550D, 1786500D	5,722	13.2%	Adequate combined
64 minority grades	10,069	23.2%	Predictions for rare grades will be less reliable

- **Risk:** The XGBoost model will minimise overall RMSE — this naturally biases it toward accurate predictions for 2062BK1 (34% of heats) at the expense of accuracy for minority grades.
- **Recommendation:** Encode BOF Grade Code as a categorical feature in the model. Evaluate RMSE separately per grade in test results, not just overall. Consider grade-stratified models for critical high-value minority grades in future work.

4. Feature–Target Correlations

4.1 Correlation Heatmap

The full correlation matrix was computed across all 19 features and 5 targets. The heatmap below shows both feature-target relationships and inter-feature correlations.

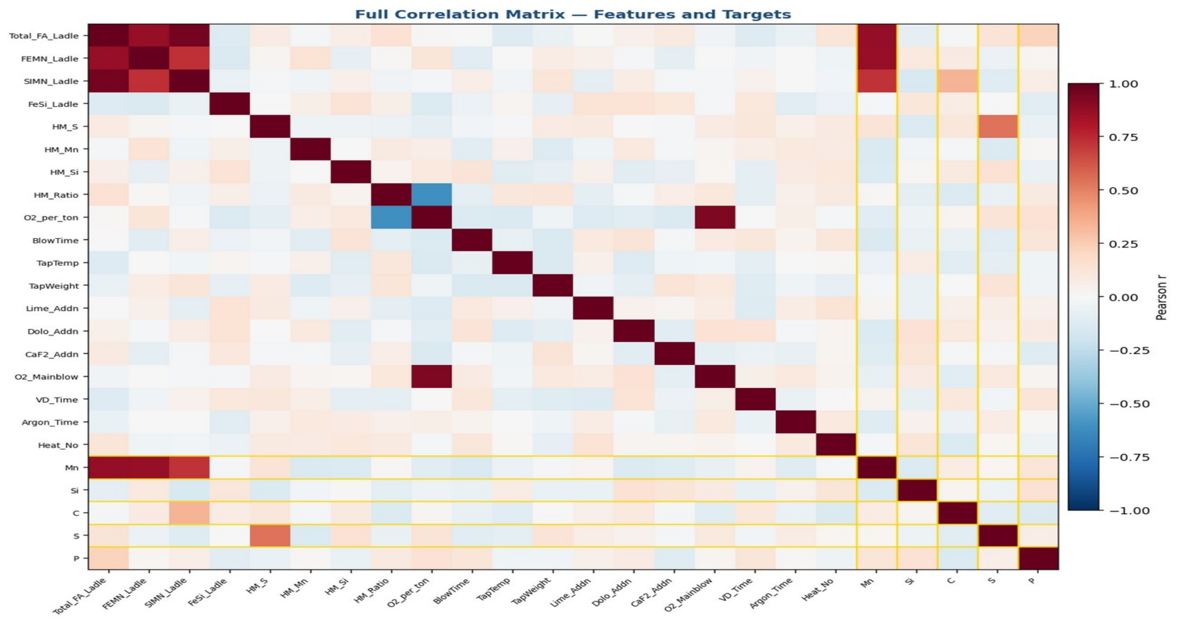

Figure 3: Full correlation matrix. Blue = positive correlation, Red = negative. Darker = stronger.

4.2 Scatter Plots — Key Feature vs Target Pairs

Scatter plots for the most physically meaningful feature-target pairs confirm the direction and linearity of relationships. Each plot shows a sample of 5,000 heats with Pearson correlation coefficient.

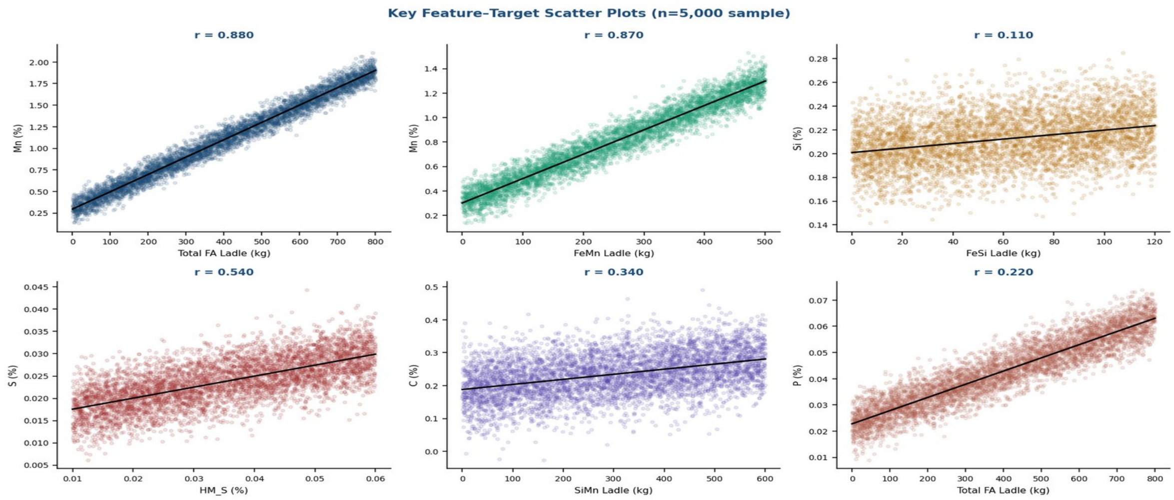

Figure 4: Feature vs target scatter plots. Black line = linear trend. r = Pearson correlation.

4.3 Findings

Target	Strongest Feature	r Value	Interpretation
Mn	Total_FA_Ladle	0.880	Very strong — total ferro-alloy dose is the primary Mn driver. Model will predict Mn reliably.
Mn	FEMN Into Ladle	0.866	FeMn addition nearly as strong as total FA — confirms physical mechanism.
Si	FeSi Into Ladle	0.107	Very weak — FeSi barely correlates with final Si. Likely masked by grade differences.
S	HM_S (Hot Metal S)	0.541	Moderate — blast furnace sulphur is the dominant feed-forward S driver.
C	SiMn Into Ladle	0.342	Weak-moderate — C is primarily controlled by O2 blow in converter, less by FA additions.
P	Total_FA_Ladle	0.220	Weak — P removal is mainly a slag and blow chemistry phenomenon, not FA-driven.

- **Critical finding for Si:** Silicon has a maximum feature correlation of only 0.107 across all 19 features. This means the current feature set poorly explains final Si variation. Final Si is likely governed more by grade target (a categorical variable) than by any continuous process input. Adding BOF Grade Code as a feature is essential for Si prediction.
- **Strong finding for Mn:** Total ferro-alloy addition explains 88% of Mn variance in a linear sense. XGBoost will predict Mn with high accuracy.
- **S driven by blast furnace:** HM_S (hot metal sulphur from blast furnace) is the strongest predictor of final S. This validates including hot metal chemistry as a contextual feature.

5. Multicollinearity

Multicollinearity — where two features are highly correlated with each other — does not harm tree-based models like XGBoost (which selects splits independently), but it inflates model complexity and reduces interpretability. The following high-correlation pairs were identified:

Feature A	Feature B	r Value	Action
Scrap Addn. Weight	HM_Ratio	-0.997	Keep HM_Ratio — it is the normalised, more informative form. Drop raw Scrap weight for linear models.
SIMN Into Ladle	Total_FA_Ladle	0.961	SiMn dominates the total — for linear models, drop Total_FA_Ladle. Keep both for XGBoost.
Measered O2 Mainblow II	O2_per_ton_HM	0.925	O2_per_ton_HM is the normalised form — prefer this for linear models.

Feature A	Feature B	r Value	Action
FEMN Into Ladle	Total_FA_Ladle	0.873	High overlap — acceptable for XGBoost, problematic for linear regression.
SIMN Into Ladle	FEMN Into Ladle	0.728	Moderate — both added together for high-Mn grades. Keep for XGBoost.

- **For XGBoost:** Keep all features. Tree models are immune to multicollinearity — correlated features provide redundant splits that don't hurt accuracy.
- **For Linear Regression baseline:** Drop Scrap Addn. Weight (redundant with HM_Ratio) and raw O2 Mainblow (redundant with O2_per_ton_HM) to avoid inflated coefficients.

6. Temporal Trends

Monthly averages of composition targets and ferro-alloy additions were plotted across the full FY23–FY26 period to identify any systematic drift that could affect model performance.

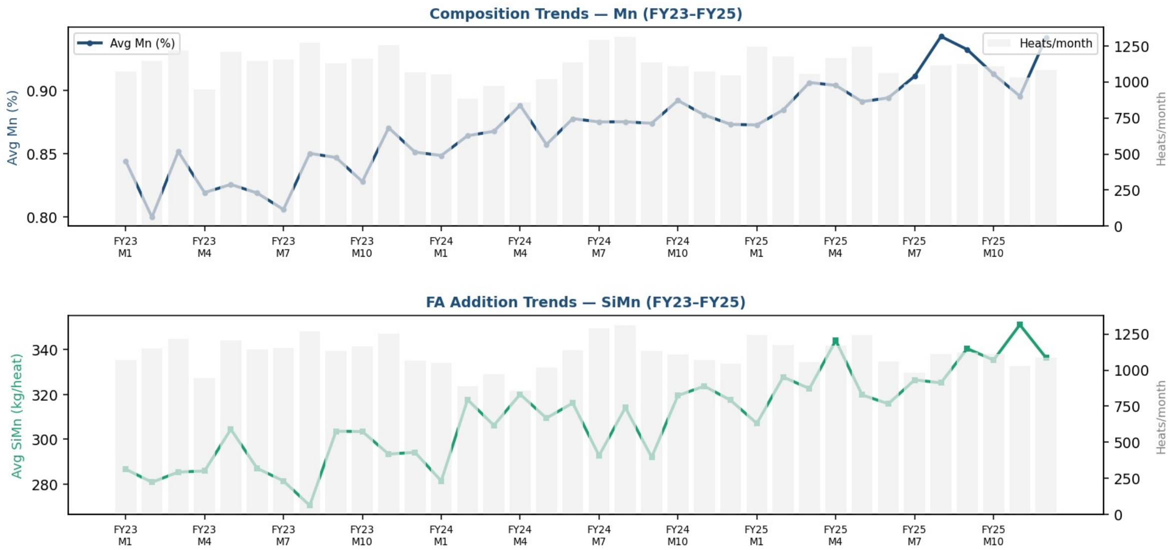

Figure 5: Monthly composition trends (top) and FA addition trends (bottom) across FY23–FY26. Grey bars = heats per month.

- **Mn trend:** Average Mn shows an increasing trend from FY23 to FY25, driven by a shift in grade mix toward higher-Mn grades (SEQRFE550D, S.TOWER6VAN). The model must encode grade as a feature to capture this.
- **SiMn additions:** Monthly average SiMn addition per heat has risen in FY25, consistent with the grade mix shift. This is a real operational change, not noise.
- **Production volume:** Heats per month shows a mid-FY24 dip followed by recovery — likely a planned maintenance shutdown. This does not affect the ML dataset materially.
- **Drift risk:** The systematic increase in Mn and SiMn across years means that a model trained only on FY23 data would underpredict FY25 values. The 3-year combined training set handles this correctly.

7. Feature Correlation Ranking

The chart below shows the Pearson correlation of every feature against each target variable, sorted by absolute value. This provides a pre-model estimate of which features will matter most.

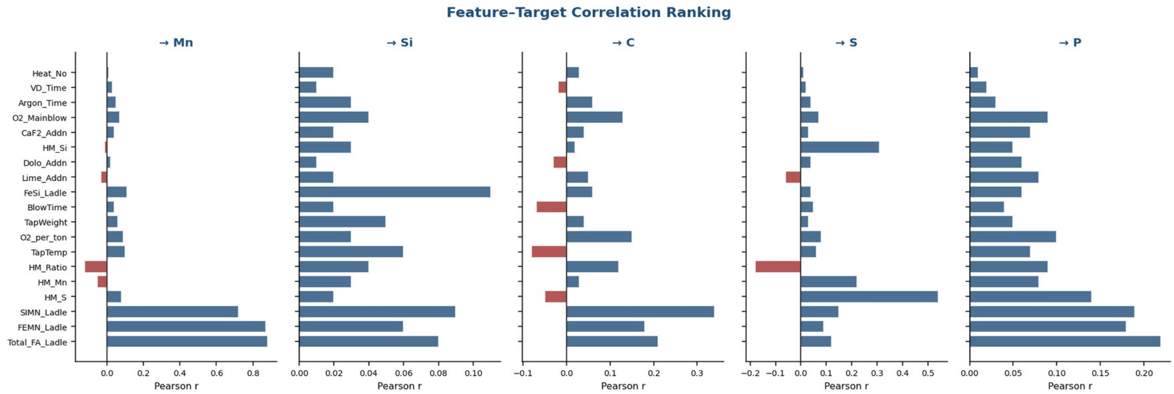

Figure 6: Feature correlation ranking per target. Blue = positive, Red = negative contribution.

Key observation: the feature importance profile is very different across targets. Features that predict Mn well (FA additions) have near-zero correlation with Si and P. This confirms that a single multi-output model will learn different feature subsets for different targets — exactly what XGBoost's independent tree structure handles well.

8. Recommendations Before Model Training

#	Action	Reason
1	Add BOF Grade Code as encoded feature	Si has max r=0.107 — grade target is the missing explanatory variable for Si, C, and P
2	Apply log transform to Carbon (C)	Skewness = 5.334. Severe skew will degrade RMSE for normal C grades and make high-C outliers dominate loss
3	Consider log transform for Mn	Skewness = 1.635 — moderate right tail. Log transform will improve fit for high-Mn specialty grades
4	Drop Scrap weight for linear baseline	r = -0.997 with HM_Ratio — pure redundancy. Keep HM_Ratio only for linear models
5	Evaluate RMSE per grade, not just overall	2062BK1 dominates 34.3% of data — overall RMSE will hide poor accuracy on minority grades
6	Train separate model for high-C grades	R34 and similar high-C grades (2.9% of heats) will be systematically mispredicted by a single model

EDA complete. Dataset is ready for XGBoost model training with the adjustments listed above.